Chapter 4

Representations of finite groups: Basic results

Recall that a **representation** of a group G over a field k is a k-vector space V together with a group homomorphism $\rho : G \to GL(V)$. As we have explained above, a representation of a group G over k is the same thing as a representation of its group algebra $k[G]$.

In this section, we begin a systematic development of representation theory of finite groups.

4.1. Maschke's theorem

Theorem 4.1.1 (Maschke)**.** *Let G be a finite group and let k be a field whose characteristic does not divide $|G|$. Then:*

(i) The algebra $k[G]$ is semisimple.

(ii) There is an isomorphism of algebras $\psi : k[G] \to \bigoplus_i \mathrm{End} V_i$ defined by $g \mapsto \bigoplus_i g|_{V_i}$, where V_i are the irreducible representations of G. In particular, this is an isomorphism of representations of G (where G acts on both sides by left multiplication). Hence, the regular representation $k[G]$ decomposes into irreducibles as $\bigoplus_i \dim(V_i)V_i$, and one has the "sum of squares formula"

$$|G| = \sum_i \dim(V_i)^2.$$